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BSIT 2 - 4

Object Oriented Programming
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Activity #1 Variables

Part A - Variable Identification Table

Variable Name	Data Type	Variable type	Stored in
name	String	Reference	Heap
grade	int	Primitive	Heap
s1	Student	Reference	Stack
s2, s3	Student	Reference	Stack

Part B - Explanation

Statement	What Happens	Memory Impact
Student s1 = new Student("John", 85);	Makes a new variable named s1 with Student data type.	A pointer named s1 will point to a memory block that contains "John" and 85.
Student s2 = new Student("Maria", 90);	Makes a new variable named s2 with student data type.	A pointer named s2 will point to a memory block that contains "Maria" and 90.
Student s3 = s2;	No new student is created.	This student s3 will point to the same memory location as s2.
s1.grade = 95;	Updates the grade of s1.	The grade that is being pointed by the pointer will change from 85 to 95.
s3.name = "Ana"	Updates the name of s3 from "Maria" to "Ana"	The name being pointed by both s2 and s3 will both change since it's the same memory.

Part C- Output

John 95

Ann 90

Ann 90